



Complete Paper Solution



EGD – Winter 2024

**Engineering Graphics and Design
(GTU)**

Q. 1 (a) Explain reducing scale. Give two practical applications of it. **(3 Marks)**



<https://youtu.be/qJ8oIUHXRQ>

Information about

1. Representative Fraction
2. Full scale
3. Reduced scale
4. Enlarged scale

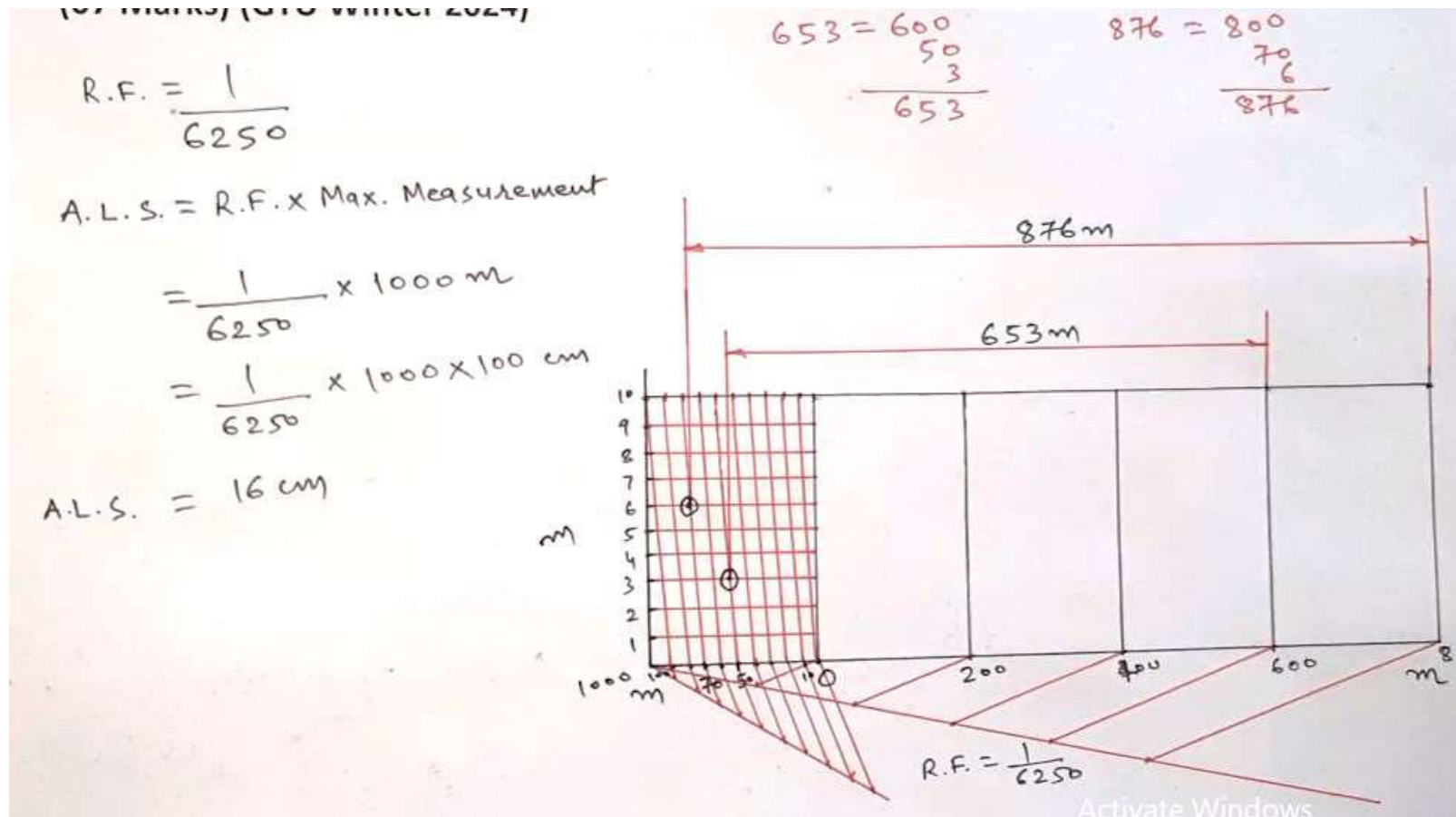
Q. 1 (b) Draw symbol of following and right use of it. 1) Cutting Plane line 2) Dimension line 3) Break line 4) Section Line **(4 Marks)**



Type	Symbol	Use
Cutting plane line		A cutting plane line is typically represented by long chain thick at ends and thin elsewhere with arrowheads at each end, and mentioned section plane name i.e. A-A, A-B etc. It is used to show cutting plane through which the object is cut.
Dimension line		A dimension line is a thin solid line and arrows symbols at the ends of dimension line to be drawn. It is used to show dimension in the drawing.
Break line		It's a short zigzag thin line used to shorten long, repetitive elements in a technical drawing like walls or fences.
Section line		A section line is a thin, continuous line drawn parallel at 45° and gap between two lines is 2 to 3 mm approximately. It is used to show the cut surfaces of an object in a drawing. They are also known as hatching lines.

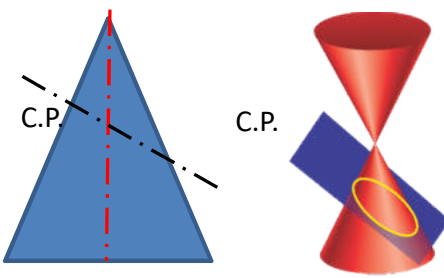


Q. 1 (c) The distance between two trees on road is 876m. Show it on a scale of 1:6250. Also show distance of 653m distance between two vehicles parked on a side of the road. (7 Marks)



Q. 2 (a) Define ellipse. What is eccentricity? What is the value of eccentricity for ellipse?
(3 Marks)

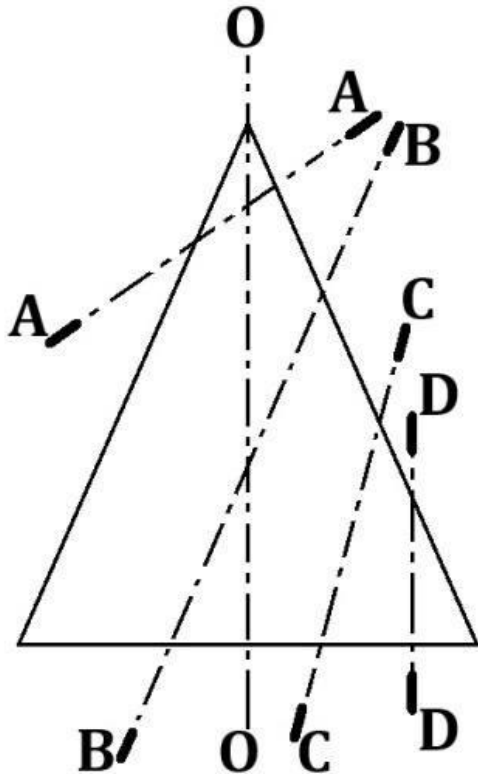


Ellipse	When the cutting plane cuts a right circular cone inclined to axis and cuts all generators, the section obtained will be ellipse.	 The diagram consists of two parts. The left part shows a blue right circular cone with a vertical dashed red line representing its axis. A black dashed line, representing a cutting plane, intersects the cone at an angle, labeled 'C.P.'. The right part shows a red double cone with a blue cutting plane intersecting it at an angle, labeled 'C.P.'. The intersection of the plane and the cone forms a yellow ellipse.
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Eccentricity (e) is the ratio of distance between vertex and focus to distance between vertex and Directrix.

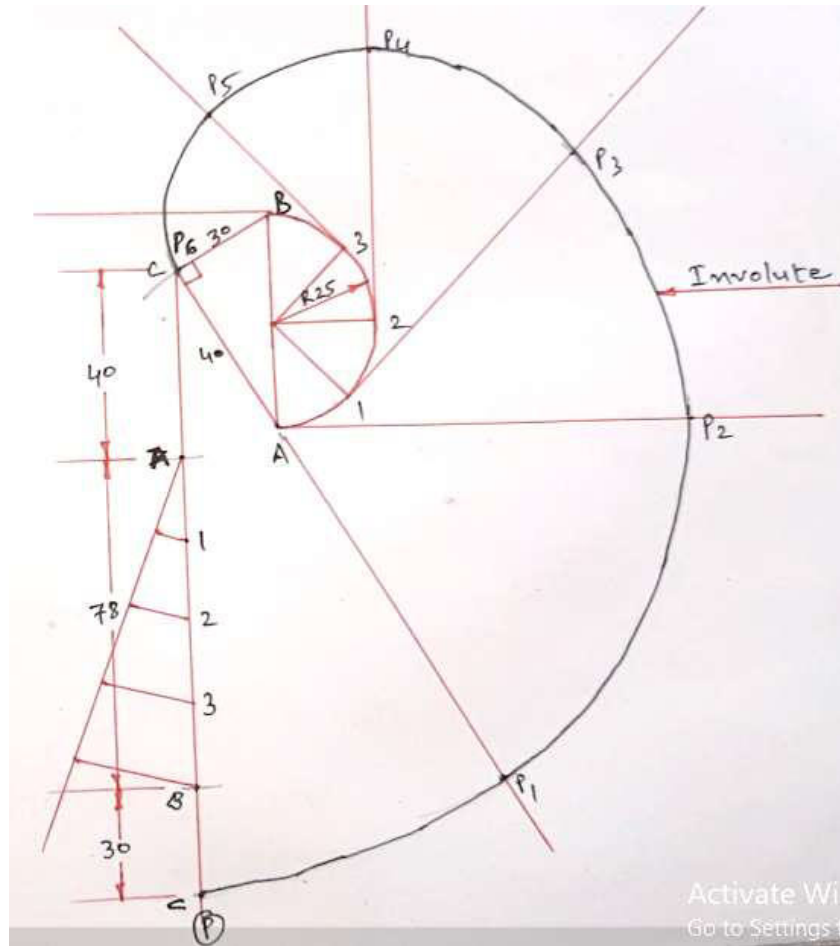
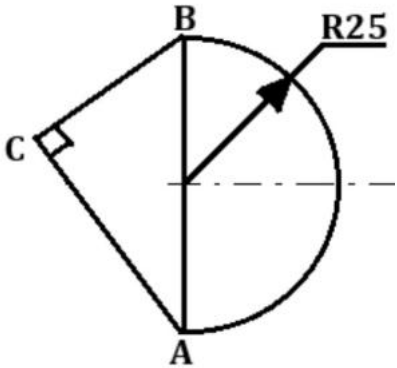
The value of eccentricity for ellipse is **less than 1**

Q. 2 (b) When we cut a cone using cutting planes as shown in figure, name the curves which are available as cut section of cone with particular cutting plane. **(4 Marks)**

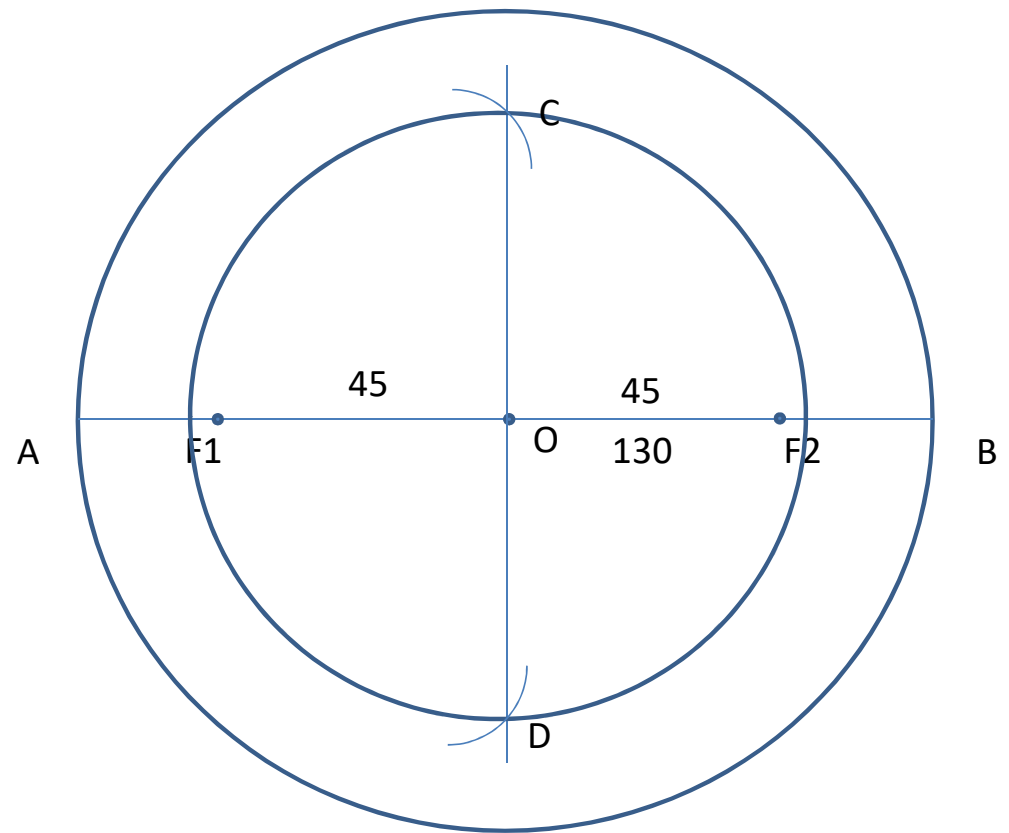
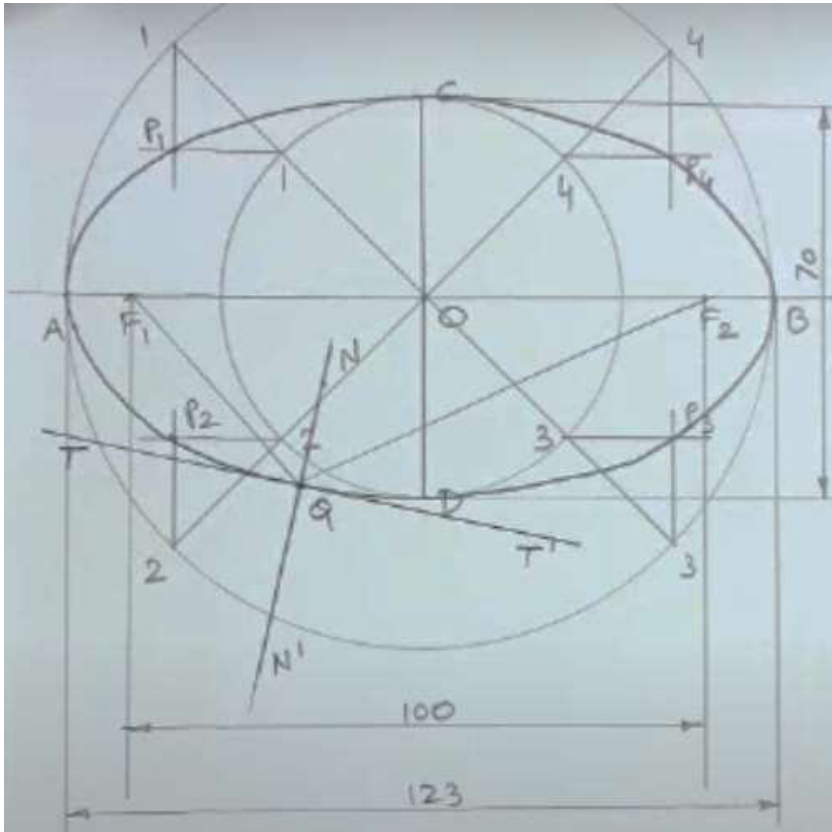


Name of Cutting Plane	Name of Curve
A-A	Ellipse
B-B	Parabola
C-C	Hyperbola
D-D	Hyperbola
O-O	Triangle

Q. 2 (c) Draw involute of a string unwound for the given figure from point C, where $AC = 40$ mm and $BC = 30$ mm **(7 Marks)**



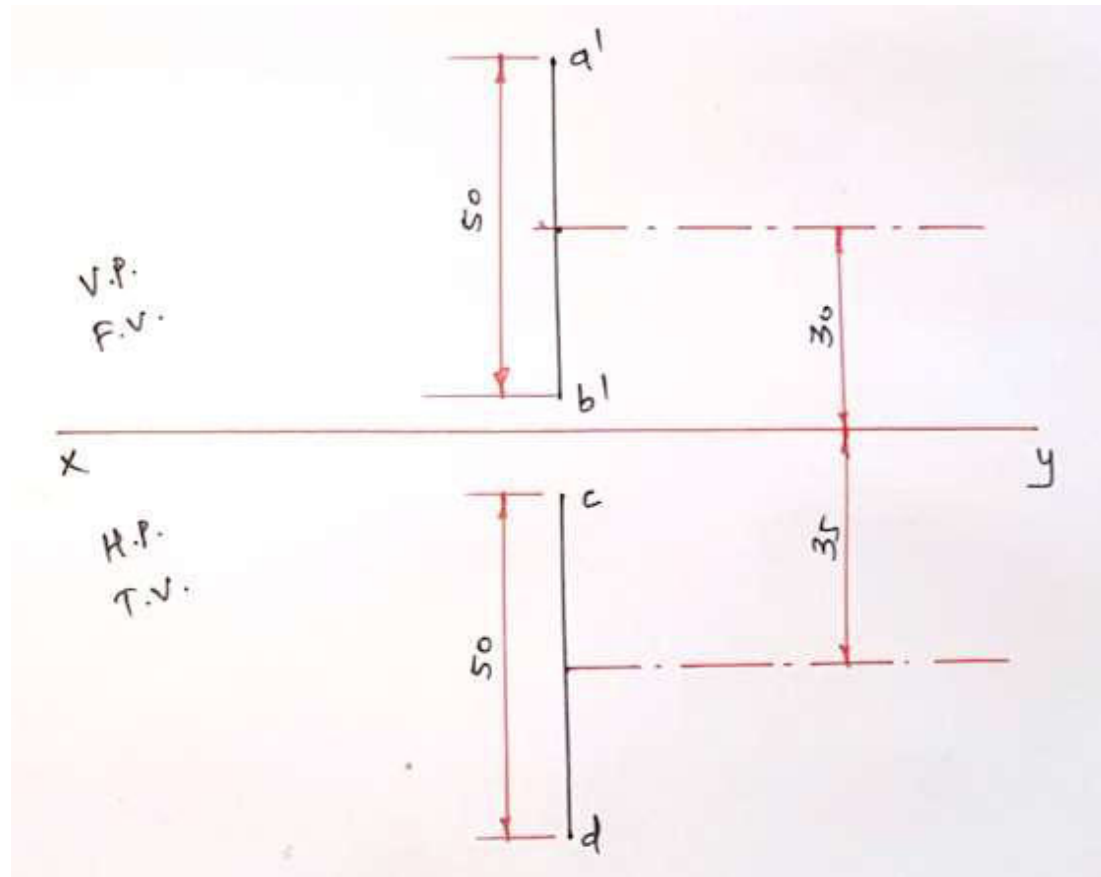
Q. 2 (c) OR Draw an ellipse using concentric circle method. Given the distance between Foci is 90mm and length of major axis is 130mm. **(7 Marks)**



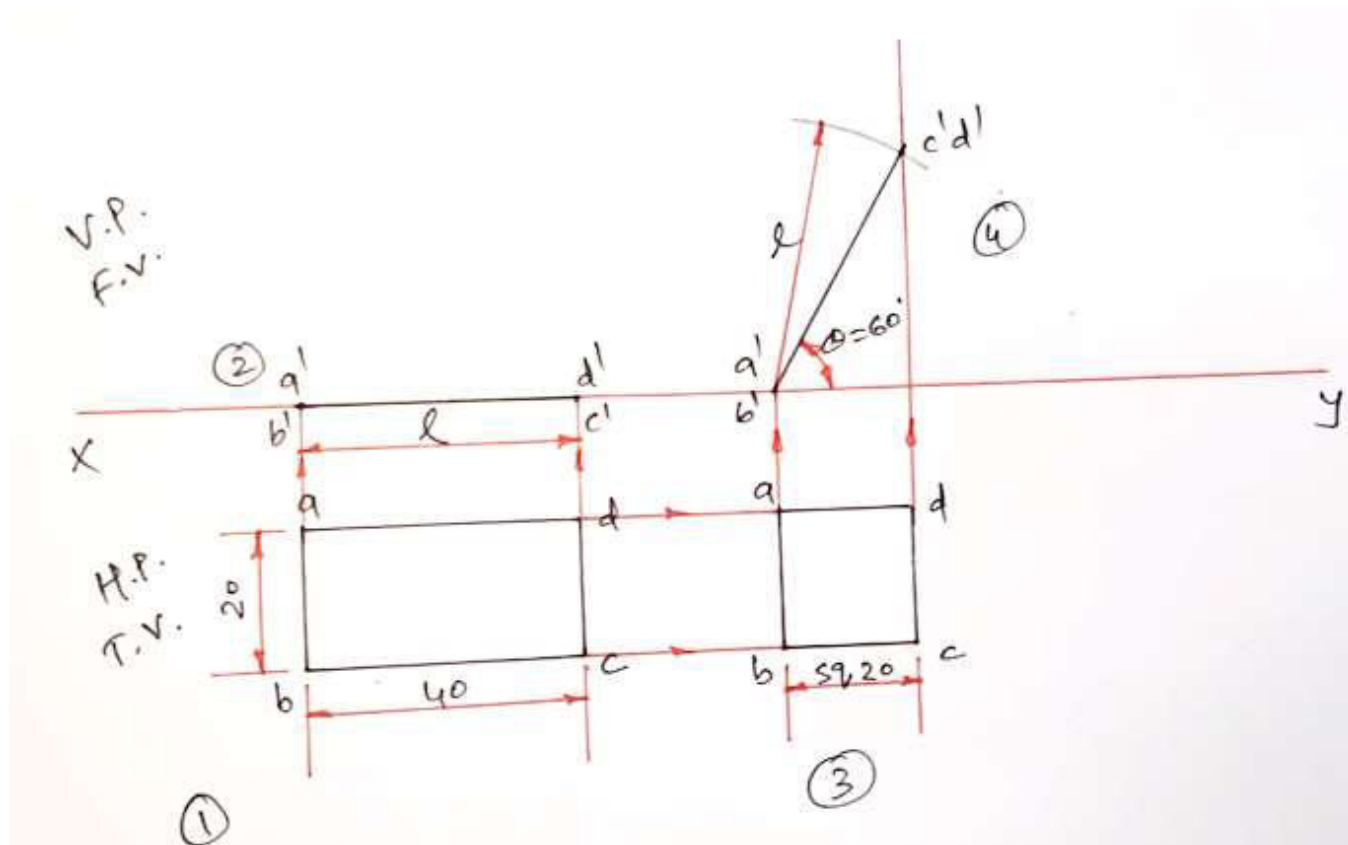
$$AO = F1C = F1D$$

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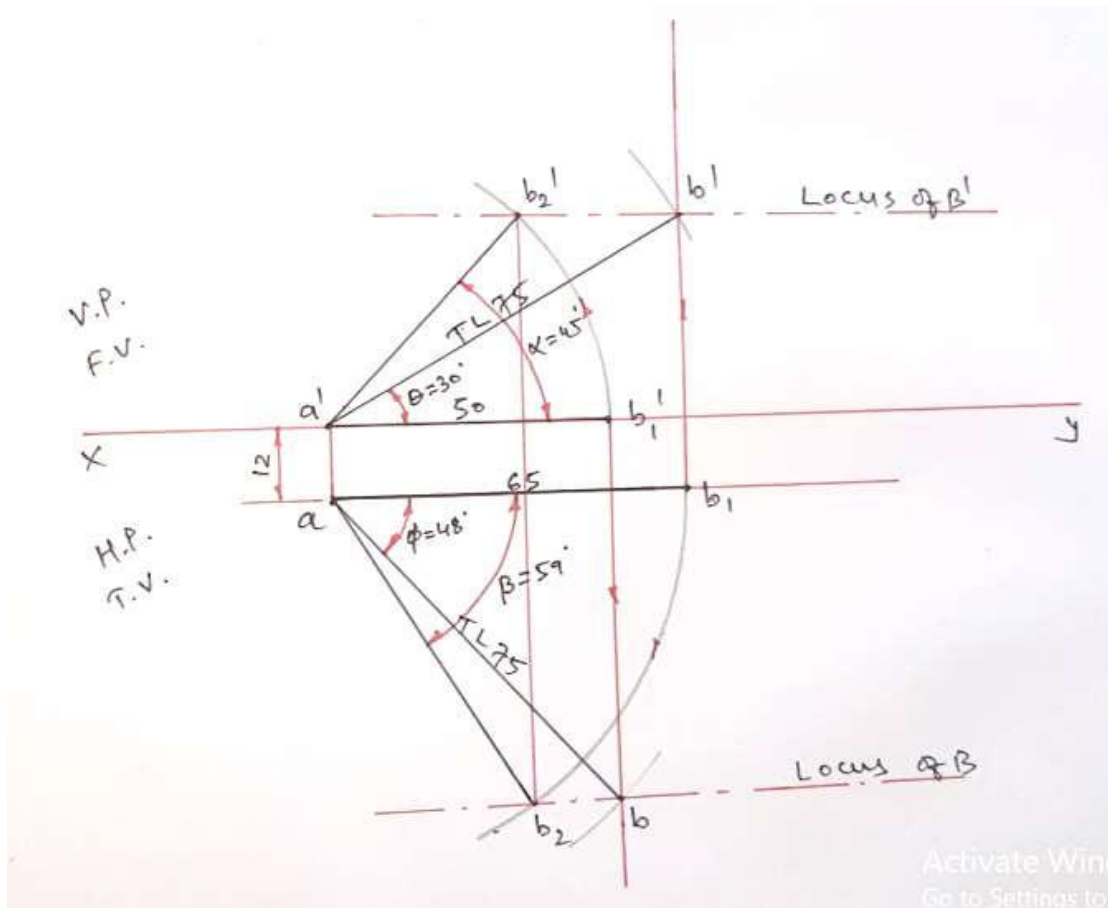
Q. 3 (a) A circular plane of 50 mm diameter is perpendicular to HP and VP. The center of plane is 30mm above HP and 35mm in-front of VP. Draw its projection. **(3 Marks)**



Q. 3 (b) A thin rectangular plate of side 40mm x 20mm has its shorter side in HP. Project its front view when its top view is a perfect square of 20mm side. **(4 Marks)**



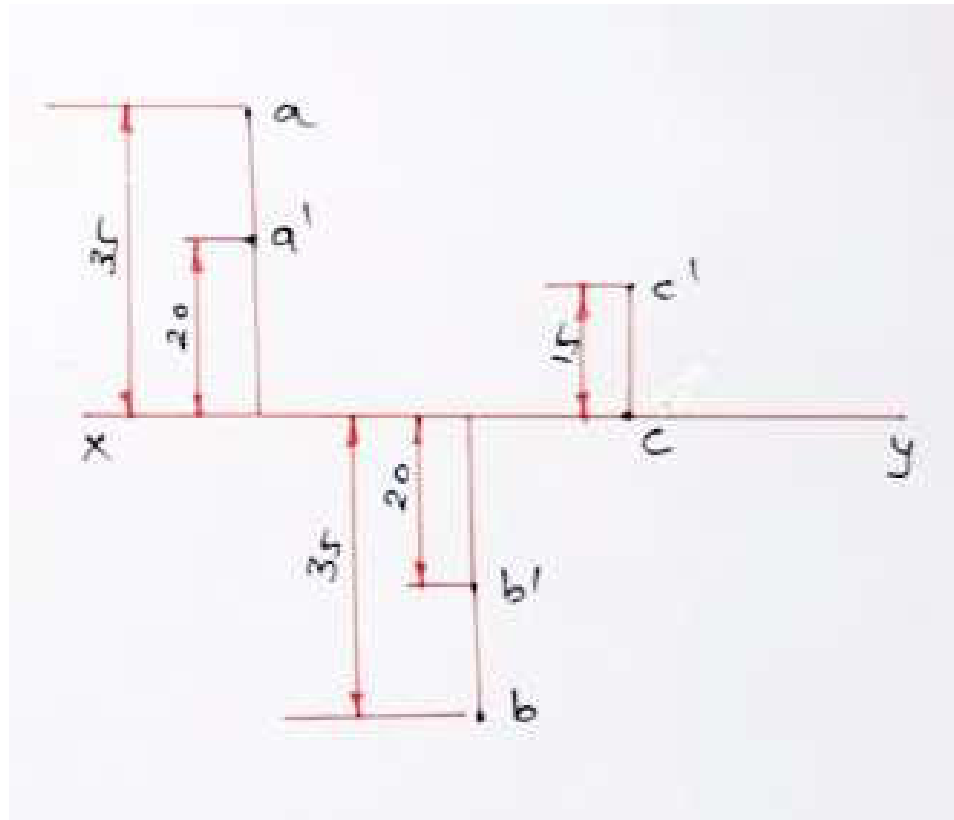
Q. 3 (c) The top view of a 75mm long line AB measures 65mm, while its front view measures 50mm. It's one end A is in HP and 12mm in front of VP. Draw the projections of AB and determine its inclination with HP and VP **(7 Marks)**



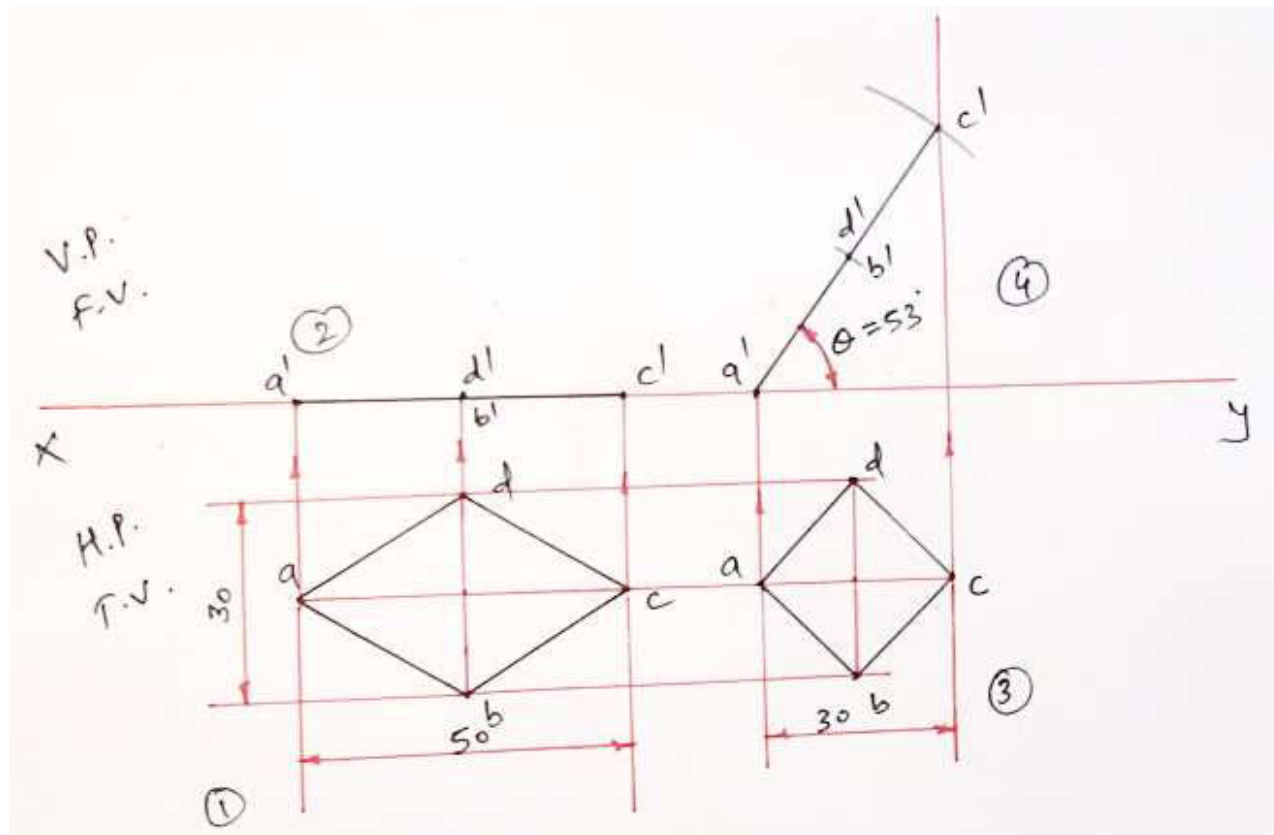


Q. 3 (a) OR Draw the projection of following points

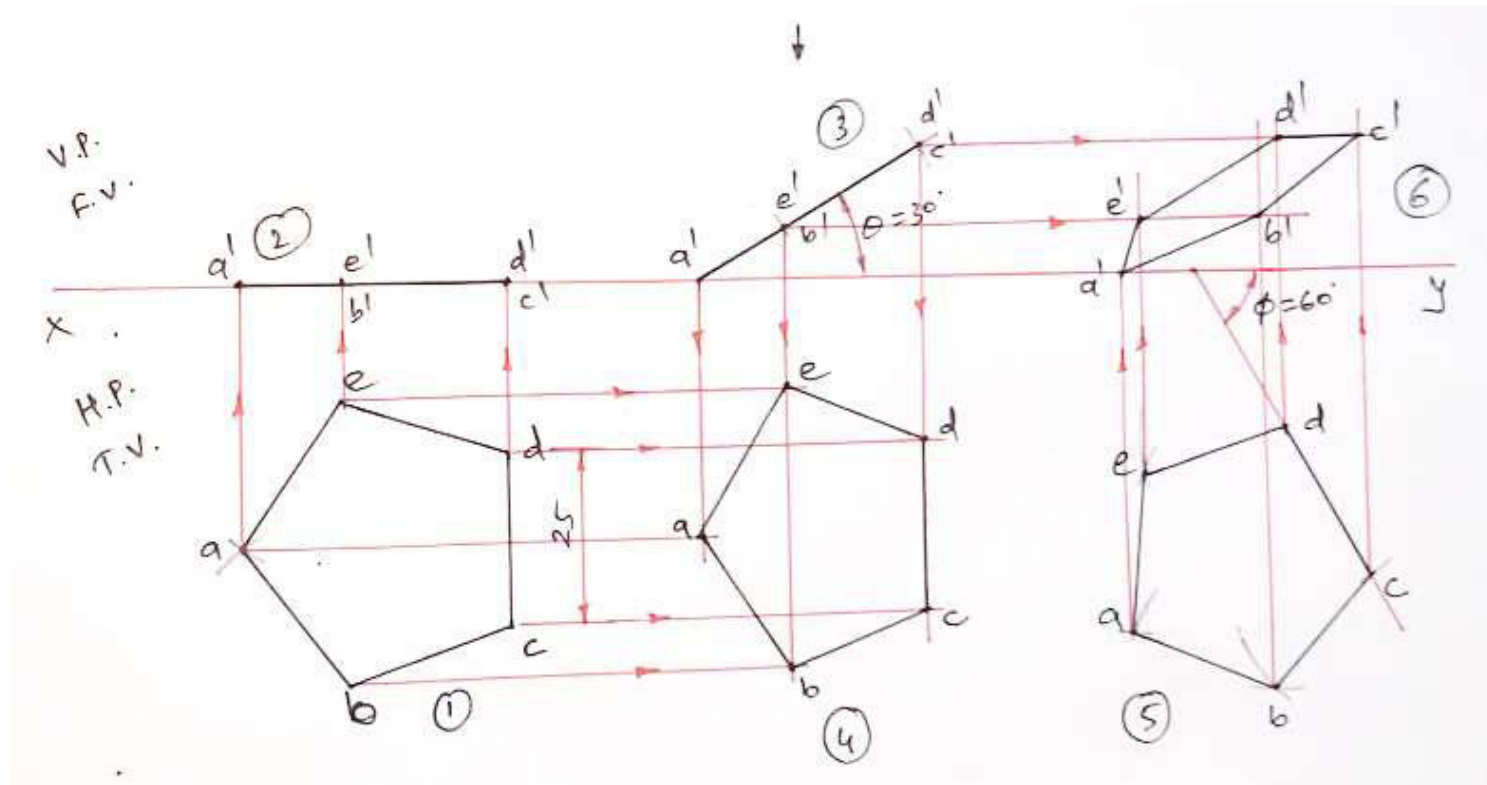
1. Point A is 20 mm above HP and 35 mm behind VP
2. Point B is 20 mm below HP and 35 mm in front of VP
3. Point C is in VP and 15 mm above HP **(3 Marks)**



Q. 3 (b) OR A rhombus of 30mm and 50mm diagonal is resting on HP on one of its corner such that it appears a square of 30mm diagonals. Find its inclination with HP.
(4 Marks)

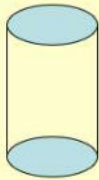
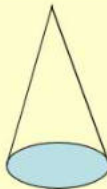
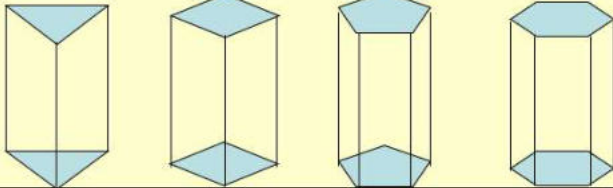
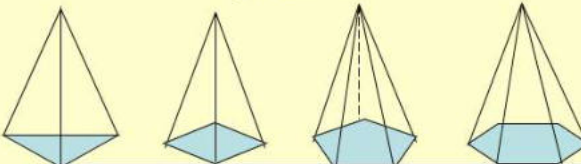
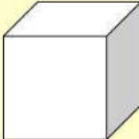
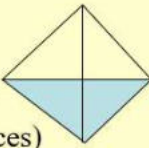


Q. 3 (c) OR A pentagonal plane having edges 25 mm is placed on one of its corners on HP such that the surface makes an angle 30° with HP and side opposite to that corner on which the plane rests inclined at 60° to VP. Draw the top and front views of the plane.
(7 Marks)



Q. 4 (a) Define solid of revolution? Give three example of solid of revolution. **(3 Marks)**



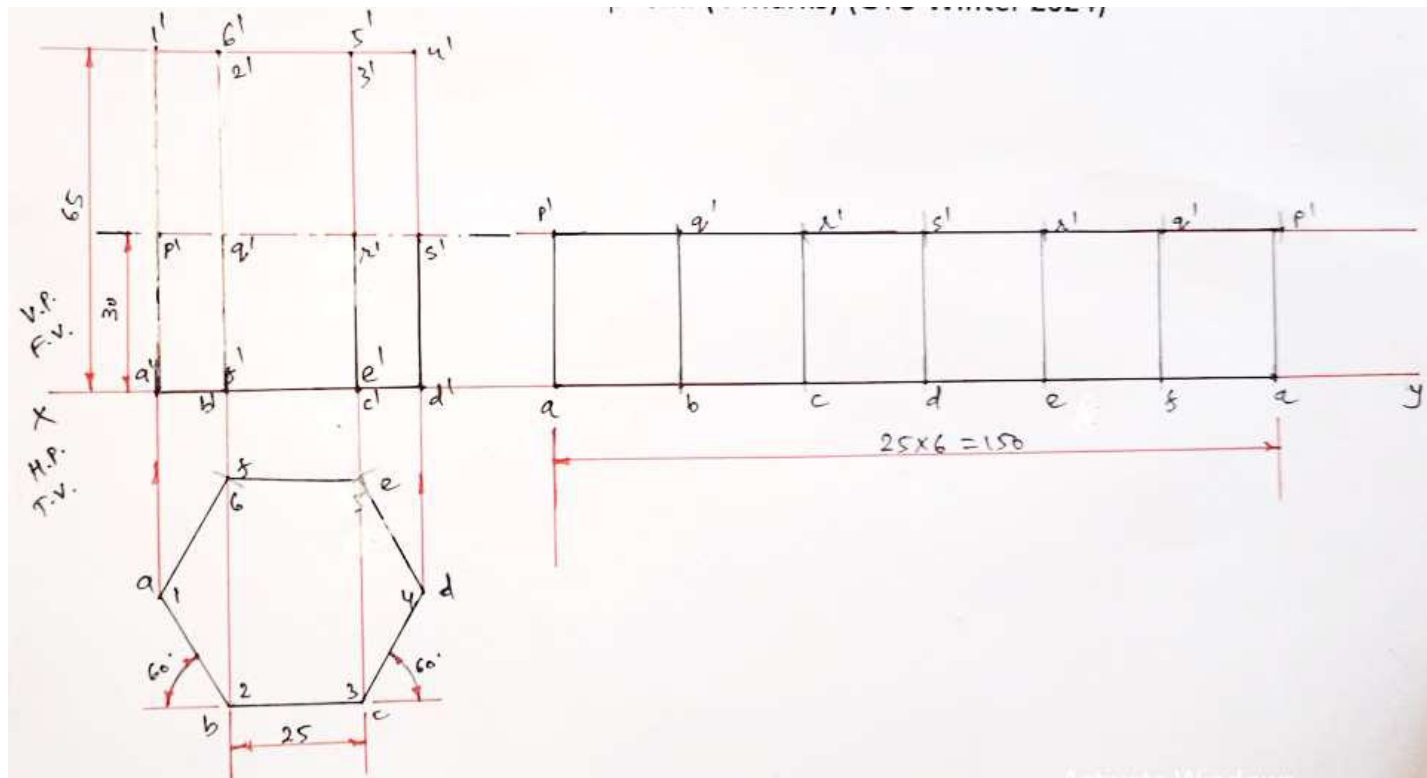
Group A				Group B			
Solids having top and base of same shape				Solids having base of some shape and just a point as a top, called apex.			
Cylinder 				Cone 			
Prisms 				Pyramids 			
Triangular	Square	Pentagonal	Hexagonal	Triangular	Square	Pentagonal	Hexagonal
Cube (A solid having six square faces) 				Tetrahedron (A solid having Four triangular faces) 			

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Video Link: <https://youtu.be/8pFcScIZ1Hs>

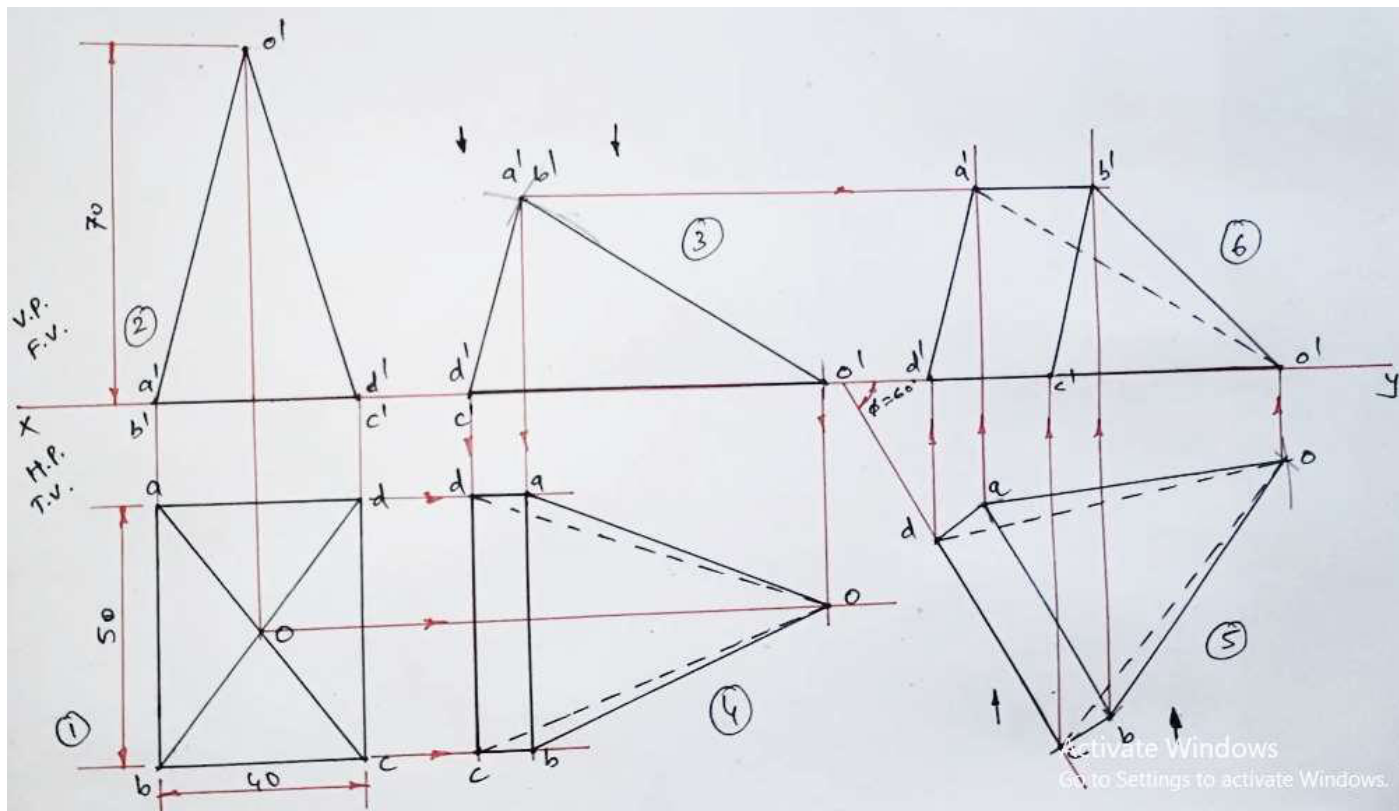


Q. 4 (b) A hexagonal prism, edge of base 25 mm and axis 65 mm long, rests with its base on H.P such that one of its rectangular faces is parallel to V.P. It is cut by a plane perpendicular to V.P and parallel to H.P and 30mm above H.P. Draw the development of the lateral surface of the truncated prism. **(4 Marks)**



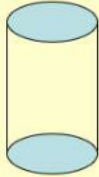
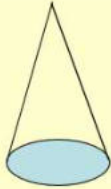
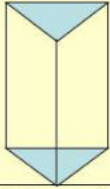
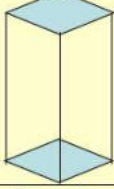
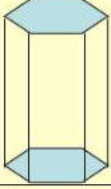
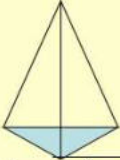
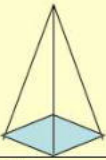
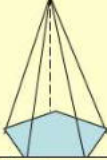
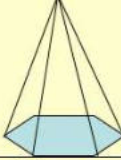
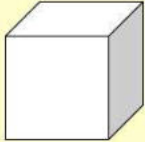
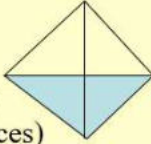


Q. 4 (c) Draw the top view and front view of a rectangular pyramid of sides of base 40 mm x 50 mm and height 70mm when it lies on one of its larger triangular faces on HP. The longer edge of the base of the triangular face lying on HP is inclined at 60° to VP in the top view with the apex of the pyramid being nearer to VP. **(7 Marks)**

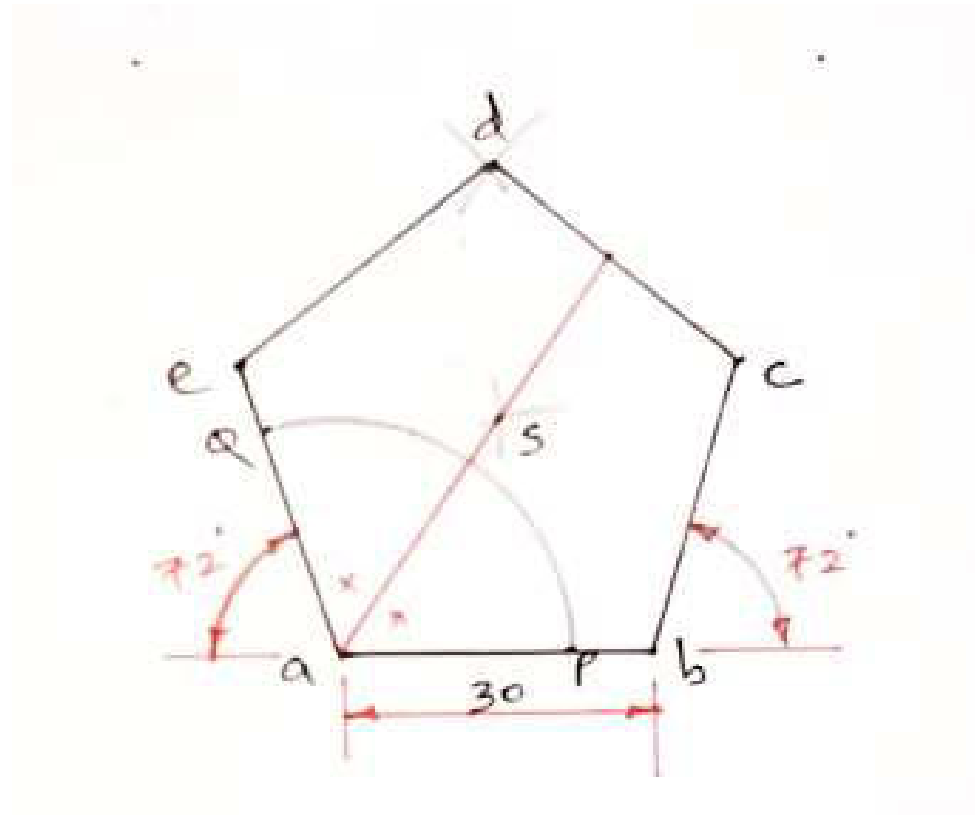




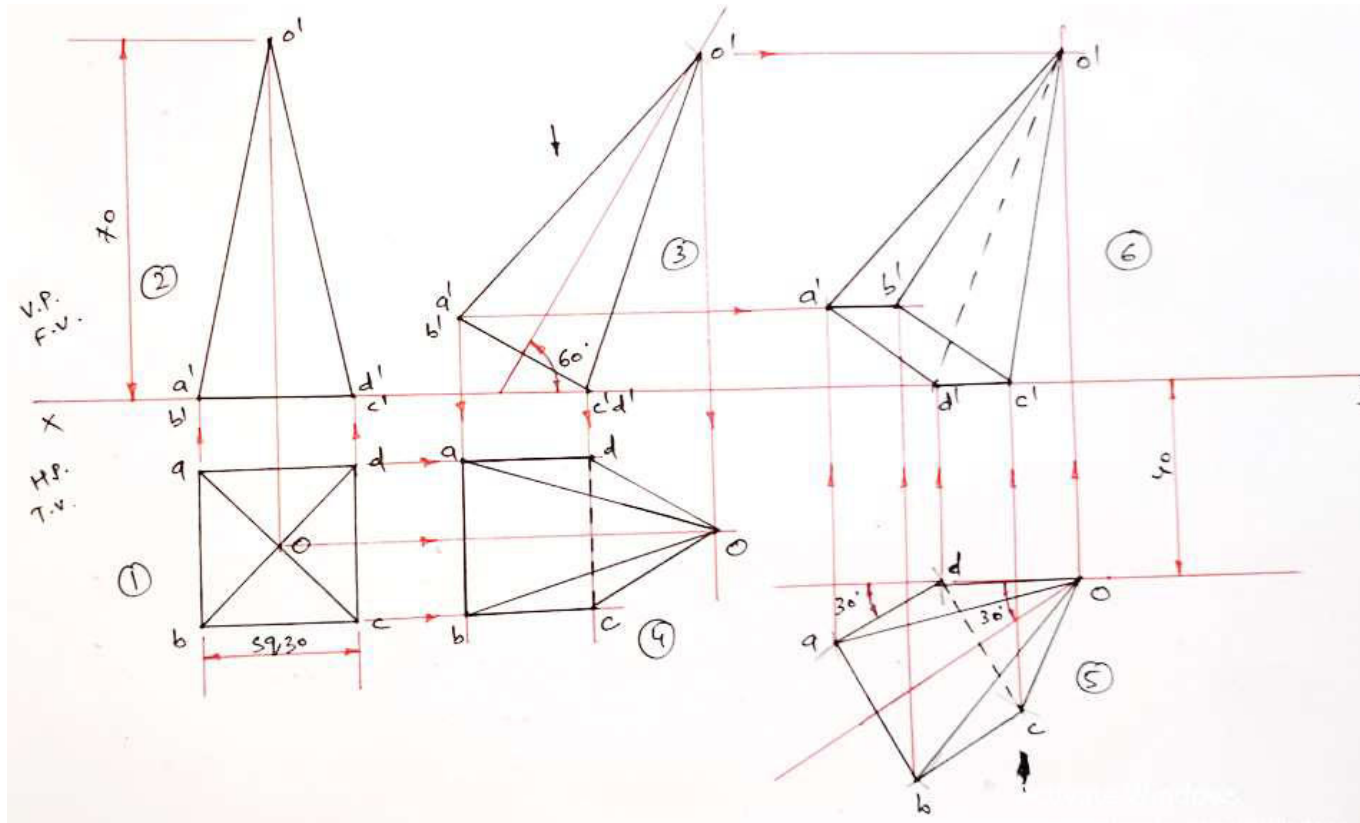
Q. 4 (a) OR Differentiate between prism and pyramid. (3 Marks)

Group A				Group B			
Solids having top and base of same shape				Solids having base of some shape and just a point as a top, called apex.			
Cylinder 				Cone 			
Prisms				Pyramids			
							
Triangular	Square	Pentagonal	Hexagonal	Triangular	Square	Pentagonal	Hexagonal
Cube (A solid having six square faces) 				Tetrahedron (A solid having Four triangular faces) 			

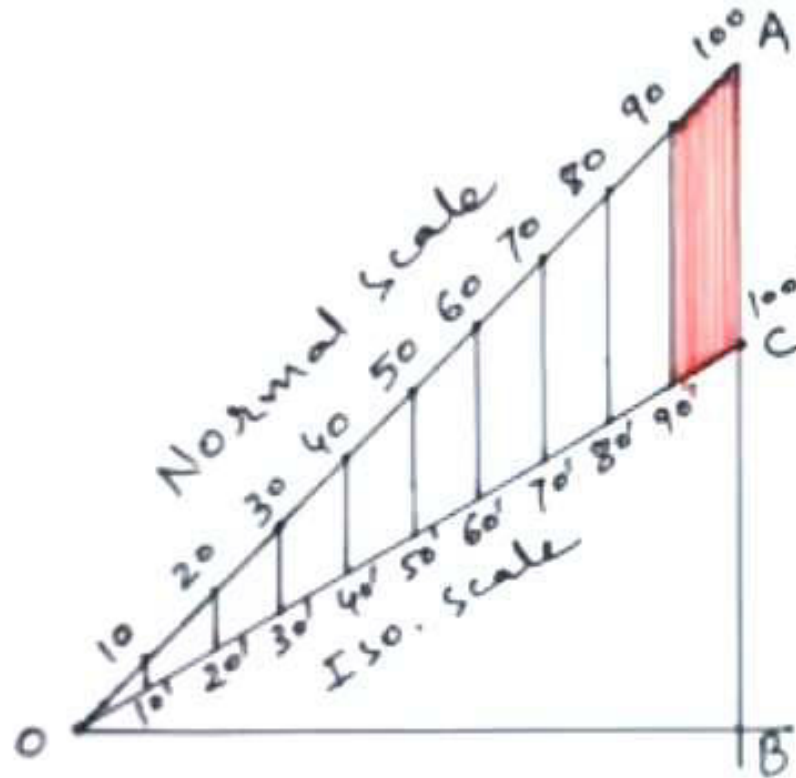
Video Link: <https://youtu.be/8pFcScIZ1Hs>



Q. 4 (c) OR A square pyramid having 30mm edge of the base and axis of 70mm long and axis parallel to VP and inclined at 60° to HP. Draw projection of pyramid if one of the edge of its base inclined at 30° to VP and apex is on HP and 40 mm away from VP. (7 Marks)

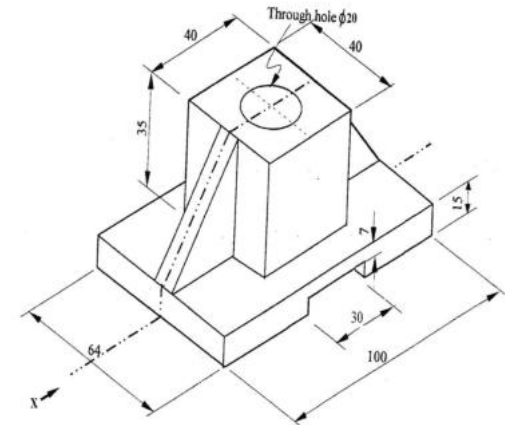
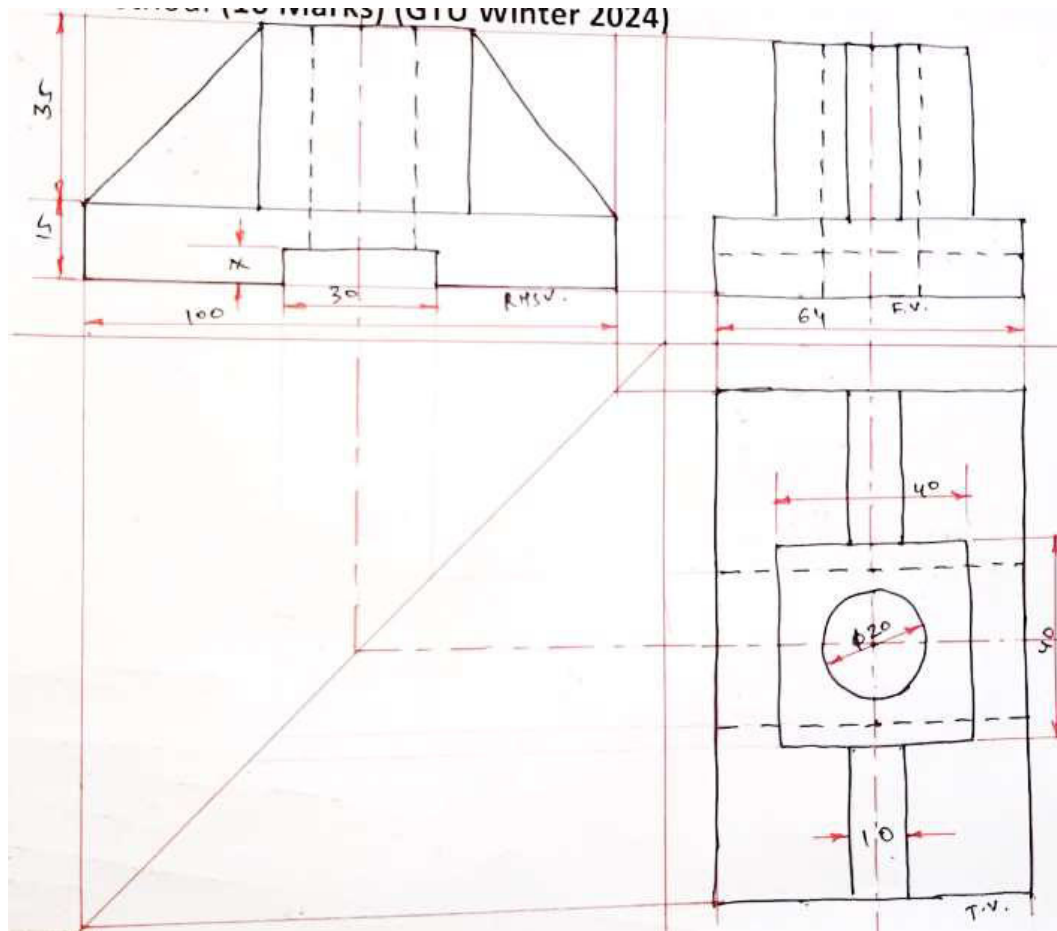


Q. 5 (a) Draw isometric scale for the line of length 100mm. Mark 53mm on it. **(4 Marks)**

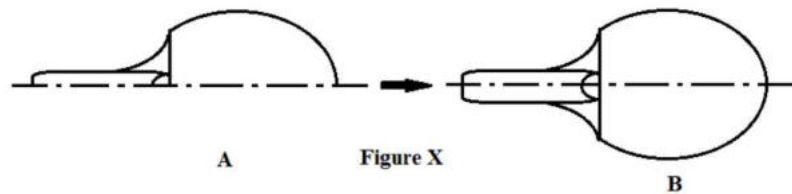


[Video Link: https://youtu.be/Bn5WVRldWm0](https://youtu.be/Bn5WVRldWm0)

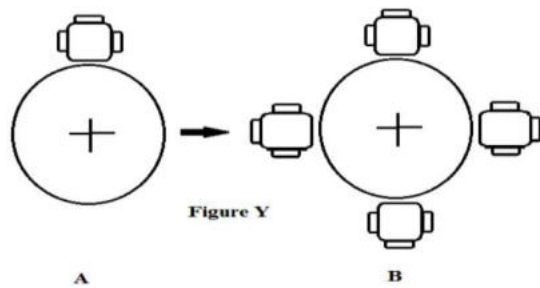
Q. 5 (b) Draw Front view, Top View and Right hand side view using 1st angle projection method. **(10 Marks)**



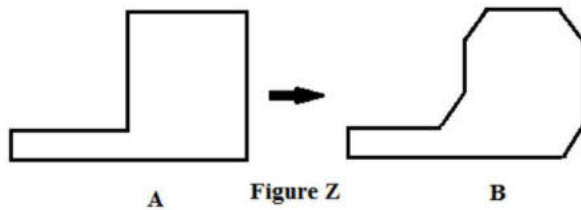
Q. 5 (a) OR Which AUTOCAD command can be used to create Part B from Part A for the following figures? **(4 Marks)**



AutoCAD Command : Mirror



AutoCAD Command : Polar Array



AutoCAD Command : Chamfer

Q. 5 (b) OR Advantages of AUTOCAD over conventional drawing. (04 Marks)



Advantages of AUTOCAD.

1. With the help of CAD highly accurate and precise drawing can be prepared in less time than manually prepared drawings.
2. Drawing instruments and hand skill is not required.
3. Drawings can be easily saved in computer and it can be modified and copied when necessary.
4. We can print the drawings on printer and plotter when required.
5. No scrap of drawings and no paper wastage.
6. Multicolor display makes visualization more effective.
7. We can increase or decrease the size of drawing to definite scale very easily.
8. Computer aided manufacturing is also possible.

Q. 5 (c) OR Draw isometric drawing of the following views. (07 Marks)

